Quantum communication and cryptography: introduction, overview on protocols, applications, challenges and importance in 5G and beyond

Safiya Salameh

1 Abstract

This paper investigates Quantum Communication (QC) and Cryptography, focusing on one of its most developed and practical applications which is Quantum Key Distribution (QKD) that exploits quantum mechanics to securely exchange encryption keys between two parties while also exploring various QKD protocols, their structures, enhancements, and practical implementation challenges besides discussing Quantum Communication applications in various aspects such as quantum optical communication. we are also going to identify the main challenges that QC suffers from scalability limitations, the challenge of integrating quantum and classical data, the development of quantum resistance cryptography which uses quantum technologies to secure systems from quantum attacks, besides one of the main issues in the quantum industry which is noise management due to quantum bits (qubits) high sensitivity to decoherence and environmental noise. The paper offers a possible solution for the coexistence of both quantum and classical data which is developing Classical-Quantum cryptographic schemes. Lastly, the paper highlights future directions for 6G using quantum communication, including quantum repeaters and long-distance quantum networks, Quantum-Enabled Meta-Surfaces (QeMS), hybrid quantum-classical architectures, and quantum-secure IoT systems.

2 Introduction

Since ancient history, civilizations aimed to enhance their communication methods due to its importance in civilising communities, starting from fire smoke, pigeons, paper and ink to sending signals. Humans have improved communication significantly. Communication shaped nations in aspects like trade, governance, cultural exchange, and science.

Humans developed their way reaching to the invention of Gutenberg's printing press and Morse's Telegraph, moving on to the widespread use of electricity after Nikola Tesla's alternating current. In the early 1800s, progress continued to telephones, radios and TVs, and eventually to the first ever computer creation in 1945.

Computers evolved rapidly over the last decade. Networking technology moved from cables to fiber optics and satellites, expanding globally with broadband and WiFi, in addition to cellular networks from 1G to 5G, making instant communication possible everywhere [1, 2]. These developments revealed new vulnerabilities regarding security.

As today's computers face a slowing pace of development due to the decay of Moore's law [3] and the Shannon limit [4], a new computing technology called *quantum computing* (QC) is emerging [5], with unique structures and abilities never experienced before in conventional computation [6] . The fundamental component of a quantum computer is the *qubit* (quantum bit) [5] [6] , which has fascinating characteristics allowing such computers to decode highly advanced encryption systems in very little time [5], creating a huge threat on classical cryptography systems.

This level of threat is beyond the reach of current classical cryptography and communication due to QC's high speed [7], as classical computing relies on computational hardness. For a quantum-level threat, quantum communication and cryptography is the best option to secure against quantum computing-based attacks [6] [8]. For example, in 2019, Google researchers built a programmable quantum computer that achieved *quantum supremacy*, completing a task in about 200 seconds that would take even the fastest supercomputer thousands of years [9].

Quantum communication is an advanced communication scheme that exploits properties of quantum mechanics such as *entanglement*, which links multiple particles with instantaneous correlations regardless of distance, and *superposition*, which allows qubits to exist in multiple states simultaneously [5] [6]. Other principles include the *no-cloning theorem*, *quantum teleportation theorem*, and *Bell's state theorem* [10] [5] [6].

Communication protocols are guidelines in telecommunication systems that enable error-free data transfer between computing systems (e.g., TCP/IP, HTTP). However, quantum communication protocols differ fundamentally, as they rely on the laws of physics rather than computational difficulty [6]. Examples include BB84 and the entanglement-based E91, typically implemented over optical fiber, free-space links, or satellites [11]. The most widely used is Quantum Key Distribution (QKD), where a quantum-secure key encrypts communication between sender and receiver. Although QKD protocols perform well in labs, real-world effectiveness depends on hardware components such as photon sources, detectors, channels, repeaters, and environmental factors, all of which affect speed, range, and security [12].

As wireless communication systems emerge, 6G can be significantly enhanced by using high-frequency microwave and terahertz bands to support extremely fast data transmission with very low latency with almost instant real time [13], but since the attempts to use such frequencies

in 5G using classical technology failed due to the higher phase noise that introduce jitter [14], as well as the propagation loss and atmospheric absorption they come with [14], with there extremely short ranges that cannot be extended using amplifiers because they would add distortion [15], and transistors cannot manage to switch fast enough to keep track with the THz level leading to burning power and introducing huge noise [15]. The key for solving this is by using quantum technologies instead, here, combined with photonic technology where lasers and resonators transmit stable-high frequencies without the need for transistors [16], beams can be bent together creating even firmer frequencies that looks like comb strands which is called the optical comb [17] [18] [19] that can push performance close to the absolute quantum noise limit [16] and the ability to use quantum communication's security is a huge advantage. Hence through quantum technologies, improve latency, throughput, capacity, scalability, energy efficiency, and especially security. Quantum communication is already implemented in the Beijing–Shanghai trunk line [20], the *Micius* satellite, and other projects [5]. However, quantum communication will not replace classical systems; instead, hybrid quantum–classical schemes will emerge, such as QKD combined with classical networks, even for quantum network nodes connected to the conventional internet [21]. This review will further discuss methods to integrate quantum technologies into communication.

3 Quantum Key Distribution and Cryptographic Implications

Today's security depends on the complexity of mathematical problems, reaching its best practice in classical cryptography. Quantum computing, using Shor's algorithm, can break RSA (based on prime factorization) and Elliptic Curve Cryptography (ECC, based on discrete logarithms) [22], which remain intractable for classical computers.

Quantum computing leverages principles such as superposition, entanglement, and quantum uncertainty [23,24]. While a classical bit is either 0 or 1, a qubit is a unit vector in a two-dimensional complex vector space, represented in equation.1 [25] as:

$$|\psi\rangle = a|0\rangle + b|1\rangle \tag{1}$$

Qubit state expressed as a superposition in the computational basis, as described in Fig. 1 [25]. Upon measurement, a qubit loses its superposition and collapses into a classical bit due to decoherence [26], making quantum computing a potent tool for hacking and quantum communication a strong defense [22].

4 Post-Quantum Cryptography

In addition to quantum communication, another promising approach is Post-Quantum Cryptography (PQC), a field focused on creating cryptographic methods immune to attacks from both traditional computers and quantum

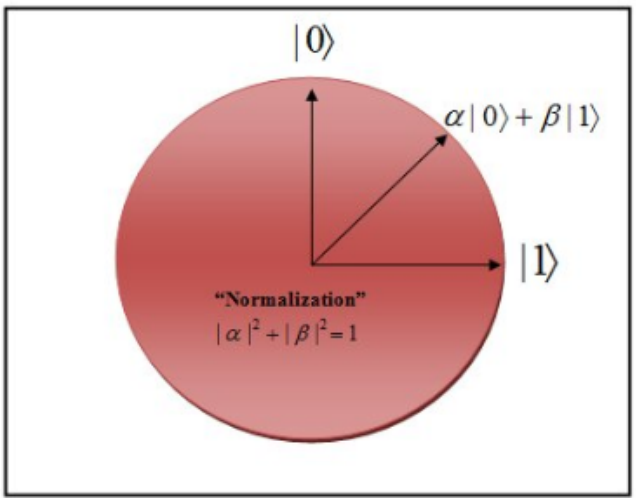

Figure 1: Normalization principle in quantum mechanics [25].

computers [27]. PQC relies on problems unsolvable by quantum computers' parallelism, which uses the superposition state of the qubit, and is implemented through PQC schemes [22].

PQC branches into multiple schemes based on the underlying mathematical problem, including lattice-based cryptography, isogeny-based cryptography, non-commutative cryptography, code-based cryptography, hash-based digital signatures, and multivariate cryptography [28]. PQC is currently being standardized by organizations such as NIST for future-proof security [28].

Principles of quantum mechanics such as quantum entanglement, and their applications like QKD, offer an alternative to classical cryptographic methods and have the potential to revolutionize communication security and efficiency [29]. Unlike classical key distribution methods, QKD relies on quantum mechanical properties [30], such as the *no-cloning theorem* and Heisenberg's uncertainty principle [31], to ensure that any eavesdropping attempt will be detectable. This is increasingly necessary given the growing threat posed by quantum computers, which can break many cryptographic protocols currently in use [22].

5 Quantum Key Distribution Protocols

QKD has undergone many developments over the years. The first and most widely known protocol is BB84, which uses photon polarization to encode key bits and detect eavesdropping [32]. Continuous-variable QKD (CV-QKD) addresses certain implementation issues by using light quadratures instead of single photons, allowing integration into standard telecom systems [33]. Other protocols address photon number splitting (PNS) attacks, such as SARG04 [34], and Distributed Phase Protocols (DPS) [35], as well as their derivatives like COW [36], RRDPS, and SNRDPS [37], and decoy-state BB84 [38].

To counter vulnerabilities in detectors and hardware, Measurement-Device-Independent QKD (MDI-QKD) [39]

and Twin-Field QKD (TF-QKD) [40] were developed for longer distances. Finally, entanglement-based protocols such as E91 [41] and Device-Independent QKD (DI-QKD) [42] aim to remove the need for trusted devices.

5.1 BB84 Protocol Basics

The BB84 is the first QKD method developed in 1984, which formed the basis of quantum cryptography, allowing people to share secret keys securely over a distance using a One-Time pad that uses a secret random key. It makes eavesdropping detectable, but a major challenge is the limitation that keys cannot be securely shared remotely [43]. The protocol is the most widely used and forms the foundation for many QKD variants.

It is based on Heisenberg's uncertainty principle, which states that it is impossible to measure or calculate exactly both the position and the momentum of an object [44].

Using the BB84 protocol, Alice can send Bob a random secret key by encoding the key in the polarization of single photons. According to the no-cloning theorem [45], Eve (an eavesdropper) cannot measure these photons and resend them to Bob without altering their state in a detectable way. However, this only holds if the quantum channel is error-free. If it is noisy, Eve's presence might not be detected [45].

Alice prepares each photon in one of four quantum states, randomly chosen from two mutually unbiased bases (rectilinear and diagonal), and sends them to Bob. The **security** relies on using **single photons**, since any measurement attempt by Eve will disturb the system and can be detected.

In practice, we **don't always send exactly one photon**. Instead, we often use **coherent states** (e.g., weak laser pulses), which behave probabilistically and may [6]:

- Send 0 photons (empty pulse),
- Send 1 photon (the intended case), or
- Occasionally send 2 or more photons (multi-photon states) **and that's where the risk lies**.

The distribution of the number of photons in a coherent state follows the **Poisson distribution**. The probability p_n of finding exactly n photons in a pulse with an average photon number $|\alpha|^2$ is given by equation.2 [6]:

$$\text{Poisson Distribution:} \quad p_n = e^{-|\alpha|^2} \cdot \frac{|\alpha|^{2n}}{n!} \tag{2}$$

This distribution arises naturally when modeling weak laser pulses [6].

Where:

$|\alpha|^2$ Average number of photons per pulse.

p_n Probability of having exactly n photons.

5.2 Distributed Phase Protocols

These protocols enhance security by using coherence across multiple rounds. In DPS [35], Alice sends coherent pulses with phase shifts of 0 or π, and Bob measures interference between consecutive pulses.

The SARG04 protocol [46] uses the same setup as BB84 but changes the basis disclosure step to resist PNS attacks [47]. The COW protocol [36,48] modulates pulse intensity instead of phase, and RRDPS/SNRDPS protocols [37] extend distributed coherence further.

5.3 Decoy state BB84

The decoy-state method was introduced by Hwang (2003), then formalized by Lo, Ma, and Chen (2005), among others. Its core principle is: instead of always using the same mean photon number μ, Alice randomly varies it between multiple values (e.g., μ_1, μ_2, μ_3). Where decoy states are lower or higher μ values that look identical to the eavesdropper and are only used for channel testing not for key generation [49].

The process: Alice prepares phase-randomized weak coherent states, which mathematically become classical mixtures of photon number (Fock) states. Coherent states like laser pulses can be described as in equation. 3 [50]:

$$|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle \tag{3}$$

Where:

- $|\alpha\rangle$: A coherent state, which is a quantum state of the electromagnetic field.
- α: A complex number that determines the amplitude and phase of the coherent state, $\alpha = |\alpha| e^{i\phi}$, and ϕ is the phase which is in the range $[0, 2\pi]$. $|\alpha|$ is the amplitude related to the average photon number.
- n: Photon number (number of photons in the Fock state) [50].
- $|n\rangle$: A Fock state which is a state with exactly n photons [50].
- $e^{-|\alpha|^2/2}$: Normalization factor to ensure the total probability is 1 [50].
- $\frac{\alpha^n}{\sqrt{n!}}$: Coefficient of each Fock state in the superposition [50].

After phase randomization, the density matrix becomes a Poisson distribution as in equation.4 [49]:

$$\rho = \sum_{n=0}^{\infty} \frac{e^{-\mu} \mu^n}{n!} |n\rangle\langle n| \tag{4}$$

Where:

- ρ: The density matrix describing the mixed state after phase randomization [50].

- $\mu = |\alpha|^2$: The mean photon number (average number of photons per pulse) [50].
- $|n\rangle\langle n|$: A projection onto the Fock state $|n\rangle$; represents a state with exactly n photons [50].
- $\frac{e^{-\mu}\mu^n}{n!}$: The Poisson distribution (probability of getting exactly n photons in a given pulse) [50].

This allows Alice and Bob to estimate how many single-photon, vacuum, and multi-photon pulses occurred, and check whether the observed statistics match expected values. If an eavesdropper performs a PNS attack, it will disturb the expected decoy statistics, which Alice and Bob can detect and abort the protocol [49].

A key result by Lim et al. (2014) showed that using three intensities is optimal for estimating the lower bound of secure single-photon detections [51]. The protocol has since been implemented in fiber-based systems over hundreds of kilometers [52], satellite links [52], and high-speed systems exceeding 10 Mbit/s [53].

5.4 Continuous variable QKD

BB84 and its derivative protocols fall under the category of discrete variable protocols (DV-QKD), as they rely on discrete observables for encoding and detecting information [54].

The original BB84 protocol uses qubits' states in a 2-dimensional Hilbert space. These qubits are typically implemented using polarization states of photons, e.g., Horizontal ($|H\rangle$) and Vertical ($|V\rangle$) or Diagonal ($|+\rangle$) and Anti-diagonal ($|-\rangle$). These polarization states are orthogonal and easily manipulated with standard optical components (beam splitters, wave plates, polarizers) [55].

CV-QKD leverages the quadrature observables of light: position-like ($\hat{q}$) and momentum-like ($\hat{p}$) operators. Light was chosen due to its ideality in speed as it travels fast, and for its nature with interacting with the environment making it better for preserving coherence. This also gives the advantage of freedom in encoding information such as polarization, phase, time-bin, frequency, spatial modes, etc [56].

Starting in the late 1990s and early 2000s, researchers like Tim Ralph started using field observables rather than just polarization modes [54]. This is where quadrature operators come in: as in quantum optics, the electromagnetic field can be described using annihilation and creation operators as in equation.5 [57]:

$$\hat{q} := \hat{a} + \hat{a}^\dagger, \quad \hat{p} := i(\hat{a} - \hat{a}^\dagger) \tag{5}$$

Here, $\hat{a}$ is the annihilation operator, and $\hat{a}^\dagger$ is the creation operator [57].

These operators satisfy the commutation relation [58]:

$$[\hat{q}, \hat{p}] = 2i \tag{6}$$

This is the commutation relation between position and momentum operators.

Now, these variables are continuous variables (CV), so instead of only encoding either 0 or 1, we can have real-numbered amplitudes [59]. Coherent states are specific types of light waves that have well-balanced uncertainty in two properties: X (amplitude) and P (phase), where the uncertainty between X and P is minimized equally, making it perfect for sending continuous information (Heisenberg's Uncertainty Principle). Its importance lies in how close it is to classical light and how easy it is to generate it using ordinary lasers [60].

In CV-QKD, instead of detecting individual photons (as in DV-QKD), we measure the wave properties of light (X and P) by using detectors like homodyne detection, which measures either X or P, and heterodyne detection, which measures both X and P simultaneously. This allows us to use affordable measurement tools like those used in regular systems and removes the need for decoy state methods [61].

This protocol works by Alice sending light signals as complex numbers where the amplitude is the real part and the phase is the imaginary part (coherent states) to Bob, who measures either just X or P (homodyne) or both X and P (heterodyne). After comparing what Bob measured to what Alice sent, they can estimate how much information Eve could have stolen, then apply error correction and privacy amplification to guarantee security [60].

The challenges in CV-QKD rely on how every photon matters if you lose it, it's as if the eavesdropper caused an error. However, in DV-QKD, photon loss is ignored. To fix this, strong error correction codes are needed, which are computationally heavy due to the infinite-dimensional Hilbert space [60].

5.5 Side channels and Hacking in Quantum Cryptography

There are main two type of attacks on QKD, the first is the attacks on the sender, one of the most known examples is Trojan Horse Attack [62]:

1. The attacker sends a pulse of light into Alice's device at a different wavelength than that of the quantum channel, to avoid detection.
2. Some of this light is reflected to the eavesdropper back from Alice's internal settings which unintentionally carries information about her chosen basis, the reflected light becomes modulated in the same way as the light used in the QKD protocol but at a different wavelength, making it difficult for the honest parties to detect the attack.

The suggested solution for this attack is using an optical isolator within the transmitter creating high loss from any counterpropagating light (two lights traveling in opposite directions along the same path or axis), thus an attacker can't inject high-power-light into the system without being detected due to damage of optical components, limiting the amount of light the attacker can send. However, attackers have attempted to damage isolators to bypass this protection [63].

The second kind of attacks is the attacks on the receiver being the most vulnerable component of a QKD system mainly because the receiver cannot rely on optical

isolators which would block legitimate signals and ruin the quantum communication, hence, the receiver is susceptible to various forms of light injection attacks, particularly when it comes to the detectors [64].

When the input state contains considerably more photons than anticipated, the behavior of single-photon detectors (SPDs), such as semiconductor SPDs can change [65]. Similar principles apply to both avalanche photodiodes (APDs), and superconducting nanowire single-photon detectors (SNSPDs), hence threshold detectors are used to notify us of any weirdness.

5.6 Twin-Field QKD (TF-QKD)

In 2018, a major improvement was introduced over earlier QKD protocols like MDI-QKD, especially in terms of how far secure communication can be maintained over optical fiber [66], a system called Twin-Field Quantum Key Distribution (TF-QKD). What previous models suffer from is linear scaling with loss which means that as the distance increases the secret key rate (the rate at which Alice and Bob can generate secure, usable bits of key per second) drops off, introducing a demanding need for repeaters to allow long distance communication. On the other hand, TF-QKD introduces what's known as square root scaling with loss that doesn't need repeaters. It works similarly to an interferometer, where Alice and Bob each control an arm. A single photon is sent to the system with Alice and Bob determining at which side of the detector the photon will end up; hence they can share a secure key without the need to trust the central measurement device, as knowing where the photon will end up (left or right of the device) is not enough for Eve to know the exact modulating settings from bits, basis, and phase. If one side is unknown, the sender and the receiver can conclude what the other's settings are, helping them to agree on a shared key, similar to the MDI-QKD [67].

TF-QKD works in a similar technique as MDI-QKD but with increased distance rate [66]. Here, two lasers' phases (they oscillate in sync to a very precise degree) are locked to each other. They modulate the laser pulses with 0 or π phase shifts, combined with the same central detection scheme [66].

Most QKD protocols suffer from rate decreasing exponentially with distance. Using the same equipment, TF-QKD allows longer secure channels. However, its challenges remain in its technical difficulty, where, for instance, we not only need frequency locking of the two lasers at Alice and Bob but also phase locking, and the need to maintain the locking across major distances is technically demanding [67].

Experiments have achieved ground-breaking distances for fiber-based quantum communication [68–70]. Recent progress in laser phase locking and in stabilizing the interferometer phase, sometimes by applying post-processing corrections instead of using active feedback, has made long-distance secure quantum communication possible [71,72], including the demonstration of over 1000 kilometers of real fiber communication [73].

5.7 Sources of Noise in QKD Systems

As QKD relies on photons and atoms, it's a highly sensitive environment where even minimal noise—caused by photon loss, scattering, and environmental disturbances can lead to disruption in photon or atom states. This results in transmission errors that raise the quantum bit error rate (QBER), often due to detector noise and background interference [74, 75]. High QBER can also open opportunities for eavesdroppers to extract data without being detected [76,77]. Other contributing factors include dark counts, timing jitter, spontaneous Raman scattering, and polarization drift [78].

5.8 Error Correction Techniques

Many methods were developed to help enhance the security of the shared key. Calderbank-Shor-Steane (CSS) codes are quantum error-correcting codes that can correct bit-flip, phase-flip errors, and both together help provide reliable key extraction even under noisy conditions [79, 80]. Entanglement purification protocols also help ensure security by using entanglement.

5.9 Entanglement Based QKD and DI QKD

Entanglement-based Quantum Key Distribution (QKD) is a fundamental technique in quantum communication, first developed in the early 1990s. The BBM92 programme, introduced in 1992 [81], found that using the quantum mechanics property quantum entanglement, a level of security matches the BB84 protocol [82]. Then E91 was a breakthrough in communication technology, as it works as follows [83]: entangled states are sent by Alice and Bob where each uses a different settings to violate CHSH inequality testing Bell's theorem; (2) those states are measured in the same computational basis and in a noiseless setup they get perfectly correlated results; (3) by Bell's theorem, if particles are entangled they cannot interact or be with anything except for the system the entanglement is created in, or else decoherence will happen so if an eavesdropper tries to access those entangled particles they will collapse into classical states losing their superposition state hence losing the data; (4) if Alice and Bob observe a strong CHSH violation it proves that the entanglement is still there and no one has dealt with the system, making it secure [83].

For many years, QKD protocols that relied on Bell test violations (like CHSH) were considered secure enough. But recent researchers found that if we close all possible loopholes in Bell tests, we can have entanglement and security of the system for sure, regardless of trusting the devices themselves, and that was the birth of Device-Independent QKD (DI-QKD) [84], where the behaviour of the devices can be random, with certain limitations mainly closing certain loopholes: Locality loophole to ensure the measurement results are independent and separated [85], Free will loophole ensures choices of measurement settings are not influenced [86], and Detection loophole to make sure lost detections can't be exploited by an at-

tacker [87]. Allowing DI-QKD to function securely even with untrusted devices [88].

That makes DI-QKD the most trusted and advanced quantum protocol represented in this paper until now but with the assurance of observing quantum principles (like Bell violations) and shutting loopholes [89]. For years, DI-QKD was seen as theoretically amazing but unrealistic practically due to its complex requirements until recently in 2022 two groups managed to implement it experimentally using trapped-ion systems to perform a loophole-free Bell test and demonstrate DI-QKD making it one of the most precise setups ever built [90], and using photonic systems with extremely high-efficiency detectors [82].

in the next page we summarize the most known QKD protocols in table.1.

6 Application Scenarios

Many applications regarding quantum communication have been a field of development. A number of major applications are mentioned below:

6.1 Quantum Optical Communication

The basic idea of quantum optical communication is to use photons (particles of light) as "flying qubits" to carry quantum information from one device to another over a network. These flying qubits are ideal for communication because they are fast, generate very little noise, and are robust to environmental effects. With the help of quantum cryptography, this type of network is highly secure. Terahertz (THz) systems can support quantum key distribution (QKD), meaning the same hardware used for THz technology can also be used in QKD systems [102]. This makes quantum optical communication a promising technology for 6G and future communication networks.

6.2 Quantum-Based Satellite Communication

Attenuation in fiber optics has always been an issue for both classical and quantum communication, however the solution for classical data transmission was to use amplifiers to send photons over long distances. On the other hand, the no-cloning theorem prevents copying the data hence a solution such as amplifiers is out of reach [103] [104]; that led to innovations of quantum repeaters which provides entanglement over short distances, and quantum memories that synchronizes teleportation over long distances. Be that as it may, quantum repeaters and quantum memories are not very applicable due to their complexity and those repeaters have been mostly theoretical and what is experimental is even city-scale. Another innovation was made but this time it proved actual real world appliance, direct photon transmission from satellites through space can transmit photons over long distances and can improve the usage of quantum repeaters and quantum memory. its only issue is quadratic beam loss [105]. Quantum channels can carry both classical and quantum information.In this system, classical information sent from planes, ships, submarines, or base stations is transmitted to a receiver through a satellite using a quantum channel. The communication process starts in the classical domain, where the input is prepared and encoded before being converted into quantum signals, similar to source and channel coding in traditional systems [106]. The data is then sent through the quantum channel. Due to possible errors, the receiver may receive incorrect bits. It then decodes the quantum signal and measures the data to recover the original message.

6.3 Terahertz Quantum Communication

THz frequencies correspond to million cycles per second. which is a speed todays communication and throughput did not successfully achieve due to multiple challenges such as higher phase noise that introduce jitter [14], the propagation loss and absorption [14], there extremely short ranges that cannot be extended using amplifiers because they would add distortion [15], and transistors inability to switch fast enough to keep track with the THz level leading to burning power and adding huge noise [15]. Quantum communication merged with photonic technology where lasers and resonators transmit stable-high frequencies without the need for transistors [16], beams can be bent together creating even firmer frequencies that looks like comb strands which is called the optical comb [17] [18] [19] that can push performance close to the absolute quantum noise limit [16] and while THz communication is high-speed, its security is limited. Quantum communication enhances security by encoding the signal using quantum technologies, then transmitting it over a classical channel for decoding [107]. Implementation requires specialized software and infrastructure that 5g lacks.

6.4 Quantum Teleportation

Quantum entanglement enables *quantum teleportation*, where the state of one quantum system can be transferred to another distant quantum system without physically moving the system [108].

6.5 Quantum Communication in 5G and 6G

6G aims to provide ultra-reliable, low-latency, high-capacity communication to support emerging applications such as the Internet of Everything (IoE) and holographic multiple-input multiple-output (MIMO) communications [109]. Quantum entanglement distribution and quantum repeaters are critical for achieving high reliability. Leveraging low Earth orbit (LEO) satellites to distribute data over 1200 km, along with quantum algorithms for network optimization, presents a promising approach to meet 6G performance demands [110, 111].Quantum communication in 5G networks is exemplified by China's hybrid QKD-5G model, where quantum signals coexist with classical data over fiber-optic links.

Table 1: Survey of Quantum Key Distribution (QKD) Protocols

Ref	Year	Protocol	Key Concept	Security Basis	Implementation Notes	Pros	Cons
[55]	1984	BB84	Two conjugate bases (Z and X)	No-cloning theorem, uncertainty	Widely implemented; basis reconciliation required	Simple, well-studied	Higher error rate than others in sim.
[91]	1991	E91	Bell inequality violations	Entanglement, Bell test	Requires entangled photon source	No need for basis disclosure	Harder to implement
[92]	1992	B92	Two non-orthogonal states	Quantum indeterminacy	Simpler state set; tested in simulations	Lowest error rate in Nurhadi's sim.	Weaker against some attacks
[93]	1992	BBM92	Entangled qubit version of BB84	Bell test + QKD	Good for long-distance; used in satellite QKD	Secure with photon-number splitting	Entanglement setup required
[92]	1998	Six-State	Adds third basis to BB84	Stronger uncertainty	More complex but more secure	Strong eavesdropping detection	Higher resource cost
[92]	2002	DPS	Differential Phase Shift	Phase-based encoding	Uses phase-encoded coherent pulses	Robust against intercept-resend	Requires phase stability
[92]	2004	SARG04	Alternative sifting method	Enhanced classical post-processing	More secure under photon-number splitting attacks	Higher robustness	More complex than BB84
[92]	2005	COW	Time-bin encoding with decoy states	Coherence + time analysis	Uses strong pulses; suited for optical fiber	Efficient long-distance comms	Lower raw security without decoys
[94]	2013	S13	Included in protocol overview	Not well specified	Survey only; implementation details unknown	Mentioned as part of survey	Details unclear
[94]	2022	Kalra & Poonia's Variant	Random bases, two-key generation	Enhanced BB84 model	Twice capacity, half error rate	Improved performance	Limited implementation so far
[95]	2023	Sasaki et al.	Uses single-photon pulses	Detectable disturbance	Security from photon interference	Clear physical security model	Photon source needed
[93]	2024	GEOQKD (Dirks et al.)	Trusted + untrusted BBM92 in orbit	Hybrid satellite QKD	41dB loss tolerance; up to 300 bps in trusted mode	Practical design for GEO QKD	Space-based setup is costly
[96]	2020s	Williams et al.	Time-bin encoding with photon pairs	Timing sync + entanglement	Sync & eavesdropping detection proven in real setup	Practical and secure	Sensitive to time-based noise
[97]	2023	Schimpf et al.	GaAs quantum dots with two-photon excitation	Bell state fidelity	Operates at 20 K to avoid degradation	Blinking-free entangled source	Needs cryogenic cooling
[98]	2023	Amer et al.	Grid with some trusted nodes	Addresses BSM & decoherence	Suggests combining trusted nodes with repeaters	More scalable grid designs	Still limited by repeater tech
[99]	2023	Ding et al.	Uses Random Forest for tuning QKD parameters	Machine learning-based tuning	Optimizes practical setups	Enhances QKD system performance	Dependent on training data
[100]	2023	Dhoha et al.	BB84 and QBC review with/without eavesdropper	Practical implementation proof	Demonstrated BB84 effectiveness even with Eve	Balanced practical insight	Limited to review
[101]	2023	Yao et al.	QKD + QRNG with entropic uncertainty theory	Trace distance, uncertainty	Proves closeness to ideal secure states	Solid theoretical foundation	Purely theoretical

The EU's OpenQKD project demonstrates QKD implementation over long distances by optimizing error correction and increasing key rates for practical applications. 6G networks will not see major advances over 5G without using quantum technologies [5].

6.6 Quantum Memory

Quantum memory is vital for enhancing long-distance quantum communication scalability [107]. However, synchronization remains time-consuming [107].

6.7 Quantum-Aided Multi-user Transmission

Quantum-aided multi-user transmission leverages quantum technologies to enhance data delivery across multiple paths. Techniques such as QSA and quantum-assisted particle swarm optimization simplify vector adjustment precoding, reduce transmission power, and significantly lower bit error rates (BER) in wireless networks [112]. Experiments show improved performance compared to classical methods [112].

6.8 Quantum Internet

Sending data through quantum devices enables the *quantum internet* [113], which has six key functionalities: quantum teleportation, quantum channels, quantum repeaters, quantum memories, QKD, and end nodes.

6.9 Underwater Quantum Communication

Most of Earth's surface is covered with water, creating opportunities for underwater communication. However, challenges exist [114]. Quantum communication underwater can solve security issues, with polarization fidelity exceeding 98% in optical-based QKD [115].

6.10 Quantum IoT

Integrating QKD with IoT ensures secure communication between billions of devices, especially with the sharp rise in the number of electronic devices in use and the trend of connecting previously offline devices to the network. This expansion is accompanied by the need to connect all these devices together in an IoT network that maintains high throughput, strong security, and most importantly high device density, along with lower latency, reduced power consumption, and overall higher performance. In order to achieve a breakthrough in these aspects, quantum technology Quantum Communication offers significant advantages. In 6G development, quantum communication is a wide gate for IoT enhanced networks [5].

6.11 Quantum Machine Learning

Machine learning (ML), a subfield of artificial intelligence (AI), that focuses on building systems that can make predictions and decisions [116]. Quantum communication and quantum computing, can enhance AI by solving previously intractable problems [117]. With the rise of 5G and 6G, larger amounts of data are being processed, however, the conventional ML methods cannot deal with massive volumes of data, due to their need for recursive processing and long training periods [118], As a result, researchers are turning to quantum computing, which offers powerful capabilities for handling complex computational problems. This gave rise to QML, a field that uses quantum mechanics to improve ML models, such as superposition and entanglement to enhance machine learning models. QML involves the development of new algorithms that often need data preprocessing steps tailored for quantum computation, like quantum supervised learning (QSL), quantum unsupervised learning (QUL), and quantum reinforcement learning (QRL) [119]. Popular QML models now include K-nearest neighbors, SVMs, evolutionary algorithms, and neural networks, which operate effectively using quantum communication applications to ensure a high level of security [116].

6.12 Industry Applications

Banking: IDQ has implemented quantum random number generators for encryption keys used in identification, secure network access, and digital signatures [116].
Credit cards: Quantum communication can protect credit card data, potentially replacing magnetic strips and EMV with nanoparticle-based security.
Government and defense: Protecting sensitive data is critical for national security, and quantum technologies offer enhanced capabilities [116].

6.13 Future of Secure Quantum Communication

Quantum technologies, exploiting unique subatomic particle behaviors, are poised to transform global communications. Governments and industries are recognizing their importance [120].

7 Challenges Regarding Quantum Communication and Cryptography

7.1 Scalability

Classical networks use amplifiers to boost signals. In quantum systems, photons cannot be amplified without destroying their state. Quantum repeaters are needed to overcome photon loss and maintain entanglement for long-distance communication [121]. Other solutions include improving entanglement distribution rates using swapping, error correction protocols [122], and extending coherence times in quantum memories [123].

7.2 Coexistence of Quantum and Classical Data

Quantum signals are fragile, often using single photons, while classical signals are strong. Sharing the same fiber can cause interference. Wavelength Division Multiplexers (WDM) allow multiple signals to coexist without compromising quantum integrity [124].

7.3 Quantum-Resistant Cryptography

Post-quantum cryptography (PQC) secures systems against quantum attacks [125]. Hybrid classical-quantum solutions combine QKD for key exchange with PQC for encryption [126].

7.4 Error Correction and Noise Management

Quantum systems are sensitive to decoherence and environmental noise. Quantum error correction codes, entanglement purification, and noise management protocols are employed to ensure reliable communication and maintain high-fidelity QKD [127].

8 Hybrid Classical-Quantum Cryptographic Schemes

Hybrid classical-quantum cryptographic schemes can produce robust systems by merging the best of both classical cryptography and QKD. As QKD provides security grounded in quantum mechanics, and classical cryptography offers high-speed data encryption and compatibility with existing networks, the resulting hybrid system is secure even against quantum threats while remaining operationally efficient [128].

Hybrid systems address QKD's bandwidth and distance limitations by combining classical cryptography. Using both technologies allows optimizing the strengths of each, complementing one another. Since QKD is theoretically immune to eavesdropping and computational attacks, combining it with classical encryption elevates overall security against both classical and quantum threats.

Classical key distribution mechanisms are prone to interception and hacking, especially over long distances. QKD in hybrid systems ensures secure, real-time key distribution with continuous monitoring for eavesdropping attempts [129].

The flexibility of quantum-classical hybrid systems allows integration with existing classical cryptographic infrastructure, enabling gradual transition without replacing legacy systems. This is crucial for sectors heavily invested in traditional cybersecurity solutions, allowing efficient implementation of quantum-resistant technologies [130].

8.1 Challenges Facing Hybrid Classical-Quantum Technology

- **Scalability and reliability of QKD systems:** Limitations of QKD affect hybrid systems in key exchange and data transmission [131].
- **Integration with existing infrastructure:** Classical communication infrastructure differs from quantum technologies, making integration challenging while maintaining efficiency and avoiding new vulnerabilities [132].
- **Cost and complexity:** QKD research and hardware are more expensive than conventional cryptography, and specialized quantum channels further increase costs [129].

9 Future Directions Using Quantum Communication in 6G

1. **Quantum Repeaters and Long-Distance Quantum Networks:** Combining satellite-based communication with terrestrial quantum repeaters extends entanglement distribution globally [116]. Micius satellite success shows potential for integrating LEO and GEO satellites with repeater nodes for continuous long-distance QKD [5]
2. **Quantum-Enabled Meta-Surfaces (QeMS):** These surfaces control quantum signals in real-time. Holographic and stacked meta-surfaces can enhance signal strength, security, and directionality, improving quantum repeaters for 6G networks [116].
3. **Hybrid Quantum-Classical Architectures:** Effective protocols are required to manage coexistence of quantum and classical signals on shared infrastructure while ensuring efficiency and security [133].
4. **Quantum Security for IoT:** 6G will need to secure massive numbers of devices per square meter. Implementing QKD for IoT ensures secure communication among these devices [134].

10 Conclusion

Over the past few years, technology evolved rapidly to make humans today almost fully reliant on it, bringing larger numbers of users, data, and devices to the internet system, besides the ultra-precise procedures from various sectors that now are technology-based. All of that and more started creating a gap between the quality of the system needed by users and the enhancements classical technologies can offer. That led to the shift towards quantum technologies, and communication technologies is no exception. Quantum communication, particularly through methods like Quantum Key Distribution (QKD), offers enhanced security and performance, making it a promising foundation for the next generation of secure, scalable

communication systems, as the paper demonstrates. After discussing QKD protocols implementation challenges, QC applications, we went on to examine Quantum Communication's challenges from qubits' high noise and environment sensitivity to scalability limitations. Add to that the very high cost and complication of integrating new quantum infrastructure, thus we need to improve quantum systems that can be used in classical environments, which also helps with the challenge of coexistence of both quantum and classical data in the same system in order for protocols to work. Currently, researchers and industries are looking for solutions every day, such as Hybrid Classical-Quantum systems that combine the best qualities from each sector. Quantum repeaters and quantum memories are a way to go for scientists to improve the restricted distance transmission scale. Another recent innovation is topological qubits, that Microsoft has proved their practical application after announcing Majorana 1 chip, which means they were able to create a new type of qubits that is not as sensitive and more controllable, which I suggest that it can be manipulated and used in Quantum Communication applications. Alongside the high reliability on the internet network, websites, and social media, those platforms have been widely known that they actually violate and sabotage users' privacy. Integrating Quantum Communications to such systems would ensure people's privacy and security, let alone companies, foundations, societies, and governments.

References

[1] "Impact of quantum computing on 5g network efficiency and data security." `https://ieeexplore.ieee.org/document/XXXXXXX`. Accessed: 2025-08-14.

[2] "On the integration and control of quantum key distribution over free-space optics and 5g networks." `https://ieeexplore.ieee.org/document/YYYYYYY`. Accessed: 2025-08-14.

[3] G. E. Moore, "Cramming more components onto integrated circuits, reprinted from electronics, volume 38, number 8, april 19, 1965, pp.114 ff.," *IEEE Solid-State Circuits Society Newsletter*, vol. 11, no. 3, pp. 33–35, 2006.

[4] M. Caleffi, K. Simonov, and A. S. Cacciapuoti, "Beyond shannon limits: Quantum communications through quantum paths," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 8, pp. 2707–2724, 2023.

[5] "Wwrf outlook 35 - quantum information technologies for 6g and beyond." `https://example.com/wwrf35quantum.pdf`. Accessed: 2025-07-12.

[6] A. Author, "Quantum cryptography: an overview of quantum key distribution," *arXiv preprint arXiv:2411.04044*, 2024.

[7] "Quantum technology for 5g/6g wireless communication systems: Applications, opportunities and challenges." `https://ieeexplore.ieee.org/document/ZZZZZZZ`. Accessed: 2025-07-12.

[8] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," *Theoretical Computer Science*, vol. 560, pp. 7–11, 1984.

[9] "The road to post-quantum cryptography." `https://transmitter.ieee.org/the-road-to-post-quantum-cryptography/?utm_source=chatgpt.com`. Accessed: 2025-08-14.

[10] "Quantum for 6g communication: A perspective." `https://example.com/quantum6gperspective.pdf`. Accessed: 2025-08-14.

[11] "Quantum communication protocols: Bb84 and e91." `https://arxiv.org/abs/1903.09051`. Accessed: 2025-07-12.

[12] "Practical challenges in quantum key distribution." `https://arxiv.org/abs/1606.05853`. Accessed: 2025-07-13.

[13] C. Han, Y. Wu, Z. Chen, and X. Wang, "Terahertz communications (teracom): Challenges and impact on 6g wireless systems," *arXiv preprint arXiv:1912.06040*, 2019. arXiv:1912.06040v2 [eess.SP].

[14] H.-J. Song and N. Lee, "Terahertz communications: Challenges in the next decade," *IEEE Transactions on Terahertz Science and Technology*, vol. 12, no. 2, pp. 105–117, 2022.

[15] T. Zugno, C. Ciochina, S. Sambhwani, P. Svedman, L. M. Pessoa, B. Chen, P. H. Lehne, M. Boban, and T. Kürner, "Use cases for terahertz communications: An industrial perspective," *arXiv preprint arXiv:2501.03823*, 2025. arXiv:2501.03823v1 [eess.SP].

[16] K. Jia, Y. Cai, X. Yi, *et al.*, "Low-noise frequency synthesis and terahertz wireless communication driven by compact turnkey kerr combs," *Nature Communications*, vol. 16, p. 6253, 2025.

[17] G. M. Macfarlane, A. S. Bell, E. Riis, and A. I. Ferguson, "Optical comb generator as an efficient short-pulse source," *Optics Letters*, vol. 21, no. 7, pp. 534–536, 1996.

[18] R. P. Kovacich, U. Sterr, and H. R. Telle, "Short-pulse properties of optical frequency comb generators," *Applied Optics*, vol. 39, no. 24, pp. 4372–4376, 2000.

[19] S. Xiao, L. Hollberg, and S. A. Diddams, "Low-noise synthesis of microwave and millimetre-wave signals with optical frequency comb generator," *Electronics Letters*, vol. 45, no. 3, pp. 170–171, 2009.

[20] "China opens 2,000-km quantum communication line." http://english.scio.gov.cn/chinavoices/2017-09/29/content_41668438.htm. Accessed: 2025-07-14.

[21] "Hybrid classical-quantum cryptographic schemes leveraging noise-resilient qkd for enhanced security." https://example.com/hybridqkd.pdf. Accessed: 2025-08-5.

[22] "Post-quantum cryptography attacks and defenses." https://arxiv.org/pdf/2202.02826. Accessed: 2025-08-14.

[23] "Explainer: What is a quantum computer?." https://medium.com/mit-technology-review/explainer-what-is-a-quantum-computer-307ab5c27acd. Accessed: 2025-08-5.

[24] "Ieee quantum computing overview." https://ieeexplore.ieee.org/document/365700. Accessed: 2025-08-5.

[25] "Quantum finite automata: survey, status and research directions." https://www.researchgate.net/publication/330576857_Quantum_finite_automata_survey_status_and_research_directions/. Accessed: 2025-08-6.

[26] "Decoherence in quantum systems." https://arxiv.org/pdf/quant-ph/0312059. Accessed: 2025-07-13.

[27] "Post-quantum cryptography: Mathematical foundations and future challenges." https://www.researchgate.net/publication/393998738_Post-Quantum_Cryptography_Mathematical_Foundations_and_Future_Challenges. Accessed: 2025-08-14.

[28] "Post-quantum cryptography." https://doi.org/10.1007/978-3-540-88702-7_1. Accessed: 2025-08-14.

[29] "Principles of quantum mechanics for 6g communication." https://arxiv.org/pdf/2502.14240. Accessed: 2025-08-14.

[30] "An introduction to practical quantum key distribution." https://www.researchgate.net/publication/349935074_An_Introduction_to_Practical_Quantum_Key_Distribution. Accessed: 2025-08-14.

[31] "Heisenberg's uncertainty principle." https://www.geeksforgeeks.org/physics/heisenberg-uncertainty-principle-definition-equation-significance/. Accessed: 2025-08-14.

[32] "Bb84 quantum key distribution protocol." https://arxiv.org/abs/2003.06557. Accessed: 2025-08-14.

[33] T. C. Ralph, "Quantum cryptography using continuous variables of single photons," *Phys. Rev. A*, vol. 61, p. 010303, 2000.

[34] V. Scarani, A. Acín, G. Ribordy, and N. Gisin, "Sarg04: A quantum key distribution protocol resistant to photon number splitting attacks," *Phys. Rev. Lett.*, vol. 92, no. 5, p. 057901, 2004.

[35] K. e. a. Inoue, "Distributed phase shift quantum key distribution," 2002.

[36] D. e. a. Stucki, "Coherent one-way quantum key distribution," *Laser & Photonics Reviews*, 2017.

[37] "Small number of random delays differential phase shift qkd." https://arxiv.org/pdf/1702.00162v1. Accessed: 2025-08-14.

[38] "Decoy-state quantum key distribution." https://arxiv.org/pdf/1302.4139. Accessed: 2025-08-14.

[39] "Measurement-device-independent quantum key distribution," *Nature Photonics*, 2014.

[40] "Twin-field quantum key distribution over long distances," *Nature*, 2018.

[41] A. K. Ekert, "Quantum cryptography based on bell's theorem," *Phys. Rev. Lett.*, vol. 67, p. 661, 1991.

[42] "Device independent quantum key distribution." https://arxiv.org/abs/1210.1810. Accessed: 2025-08-14.

[43] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," *Theoretical Computer Science*, 2020. Originally presented in 1984.

[44] W. Heisenberg, "Über den anschaulichen inhalt der quantentheoretischen kinematik und mechanik," *Zeitschrift für Physik*, 1927.

[45] W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned," *Nature*, 1982.

[46] "Sarg04 qkd protocol." https://arxiv.org/pdf/quant-ph/0211131. Accessed: 2025-08-14.

[47] "Enhanced sarg04 protocol implementation." https://arxiv.org/pdf/2411.04044. Accessed: 2025-08-14.

[48] "The security of practical quantum key distribution." https://www.researchgate.net/publication/1912972_The_Security_of_Practical_Quantum_Key_Distribution. Accessed: 2025-08-14.

[49] H.-K. Lo, X. Ma, and K. Chen, "Practical decoy state for quantum key distribution," *arXiv preprint arXiv:1302.4139*, 2013.

[50] D. Rosenberg *et al.*, "Experimental quantum key distribution with decoy states," *Physical Review Letters*, vol. 94, no. 23, p. 230504, 2005.

[51] C. C. W. Lim, M. Curty, N. Walenta, F. Xu, and H. Zbinden, "Three-intensity decoy-state method for quantum key distribution," *Physical Review A*, vol. 89, no. 2, p. 022307, 2014.

[52] S.-K. Liao *et al.*, "Satellite-to-ground quantum key distribution," *Nature*, vol. 549, no. 7670, pp. 43–47, 2017.

[53] F. Grunenfelder *et al.*, "High-speed quantum key distribution exceeding 10 mbit/s," *npj Quantum Information*, vol. 9, no. 1, pp. 1–7, 2023.

[54] A. K. Ekert, "Quantum cryptography based on bell's theorem," *Physical Review A*, vol. 61, no. 1, p. 010303, 2000.

[55] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," *Theoretical Computer Science*, 1984. Presented at IEEE International Conference on Computers, Systems and Signal Processing, Bangalore, India.

[56] S. L. Braunstein and P. van Loock, "Colloquium: Quantum information with continuous variables," *Reviews of Modern Physics*, vol. 84, no. 2, pp. 621–669, 2012.

[57] R. Bavdekar, E. Jayant Chopde, A. Agrawal, A. Bhatia, and K. Tiwari, "Post quantum cryptography: A review of techniques, challenges and standardizations," in *2023 International Conference on Information Networking (ICOIN)*, pp. 146–151, 2023.

[58] R. Loudon, *The Quantum Theory of Light*. Oxford University Press, 09 2000.

[59] C. Weedbrook *et al.*, "Continuous-variable quantum information: Gaussian states and beyond," *Reviews of Modern Physics*, vol. 77, no. 2, pp. 513–578, 2005.

[60] T. C. Ralph, "Quantum key distribution using continuous variables of single photons," *Physical Review A*, vol. 62, no. 6, p. 062306, 2000.

[61] F. Grosshans and P. Grangier, "Quantum key distribution using gaussian-modulated coherent states," *Physical Review Letters*, vol. 88, no. 5, p. 057902, 2002.

[62] "Trojan horse attack on qkd." `https://doi.org/10.48550/arXiv.1704.07749`. Accessed: 2025-08-14.

[63] "Damage of optical isolators to bypass qkd protection," *Phys. Rev. A*, vol. 94, p. 030302, 2016.

[64] "Attacks on the receiver in qkd systems," *Phys. Rev. Lett.*, vol. 112, p. 070503, 2014.

[65] "Invited review article: Single-photon sources and detectors." `https://www.researchgate.net/publication/51537407_Invited_Review_Article_Single-photon_sources_and_detectors`. Accessed: 2025-08-14.

[66] "Twin-field quantum key distribution over long distances," *Nature*, 2018.

[67] "Quantum optics: a toolbox for photonic quantum technologies," *arXiv preprint arXiv:2202.02826*, 2022.

[68] "Long-distance fiber-based quantum communication," *Nature Photonics*, 2020.

[69] "High-rate fiber-based qkd experiments," *Nature Photonics*, 2019.

[70] "Experimental demonstration of long-distance qkd," *Phys. Rev. X*, vol. 9, p. 021046, 2019.

[71] "Post-processing phase stabilization in qkd systems," *Nature Photonics*, 2021.

[72] "Long-distance fiber qkd with stabilized interferometers," *Nature Photonics*, 2021.

[73] "Over 1000 km fiber-based quantum communication demonstration," *Phys. Rev. Lett.*, vol. 130, p. 210801, 2023.

[74] S. Khanna *et al.*, "Quantum ber estimation modelling and analysis for satellite-based quantum key distribution scenarios," *IET Quantum Communication*, vol. 5, no. 1, pp. 1–12, 2024.

[75] M. Mlejnek *et al.*, "Reducing spontaneous raman scattering noise in high quantum bit rate qkd systems over optical fiber," *arXiv preprint arXiv:1712.05891*, 2017.

[76] S. Nauerth *et al.*, "Estimating the impact of light pollution on quantum communication between qeyssat and canadian quantum ground station sites," *EPJ Quantum Technology*, vol. 12, pp. 1–14, 2025.

[77] A. Carrasco-Casado *et al.*, "Feasibility of quantum key distribution from high-altitude platforms," *Quantum Science and Technology*, vol. 6, no. 3, p. 035025, 2021.

[78] J. Lee, M. Kim, and H. Park, "Optimizing continuous-wave pumped entanglement-based qkd in noisy environment," *arXiv preprint arXiv:2502.15059*, 2025.

[79] "Good quantum error-correcting codes exist," *Phys. Rev. A*, 1996.

[80] "Simple quantum error correcting codes," *Phys. Rev. A*, 1996.

[81] S. L. Braunstein, A. Mann, and M. Revzen, "Proposed test of bell's inequality with efficient detectors," *Physical Review Letters*, vol. 68, no. 5, pp. 557–560, 1992.

[82] W.-Z. Zhang *et al.*, "Experimental device-independent quantum key distribution," *Nature*, vol. 607, no. 7917, pp. 687–691, 2022.

[83] A. K. Ekert, "Quantum cryptography based on bell's theorem," *Physical Review Letters*, vol. 67, no. 6, pp. 661–663, 1991.

[84] A. Acín *et al.*, "Device-independent security of quantum cryptography against collective attacks," *arXiv preprint arXiv:1210.1810*, 2012.

[85] V. Scarani *et al.*, "Randomness versus nonlocality and entanglement," *International Journal of Quantum Information*, vol. 8, no. 4, pp. 525–540, 2010.

[86] M. J. W. Hall, "Local deterministic model of singlet state correlations based on relaxing measurement independence," *Physical Review Letters*, vol. 105, no. 25, p. 250404, 2010.

[87] L. Lydersen, C. Wiechers, C. Wittmann, D. Elser, J. Skaar, and V. Makarov, "Hacking commercial quantum cryptography systems by tailored bright illumination," *Nature Photonics*, vol. 4, no. 10, pp. 686–689, 2010.

[88] S. Pironio *et al.*, "Security of device-independent quantum key distribution in the bounded-quantum-storage model," *Physical Review Letters*, vol. 108, no. 13, p. 130503, 2012.

[89] H.-K. Lo, M. Curty, and B. Qi, "Measurement-device-independent quantum key distribution," *Physical Review Letters*, vol. 108, no. 13, p. 130503, 2012.

[90] D. P. Nadlinger *et al.*, "Experimental quantum key distribution certified by bell's theorem," *Nature*, vol. 607, no. 7917, pp. 682–686, 2022.

[91] A. K. Ekert, "Quantum cryptography based on bell's theorem," *Physical Review Letters*, vol. 67, no. 6, pp. 661–663, 1991.

[92] N. Nurhadi *et al.*, "Quantum key distribution (qkd) protocols: A survey," in *IEEE Conference Publication*, Various years depending on protocol.

[93] D. et al., "Geoqkd: Quantum key distribution from a geostationary satellite," *TU Delft Research Portal*, 2024.

[94] M. Kalra and R. C. Poonia, "Design a new protocol and compare with bb84 protocol for quantum key distribution," in *Proceedings of SocProS 2017, Volume 2*, 2022.

[95] H. Takesue, T. Sasaki, K. Tamaki, and M. Koashi, "Experimental quantum key distribution without monitoring signal disturbance," *Nature Photonics*, vol. 9, p. 827–831, Sept. 2015.

[96] W. et al., "Implementation of quantum key distribution and quantum clock synchronization via time bin encoding." Preprint, 2020. Available on request or institutional archive.

[97] H. Schimpf and et al., "Entanglement-based quantum key distribution with a blinking-free quantum dot operated at a temperature up to 20 k," *arXiv preprint arXiv:2108.13618*, 2023.

[98] O. Amer and et al., "Efficient routing for quantum key distribution networks," *arXiv preprint arXiv:2005.12404*, 2023.

[99] X. Ding and et al., "Predicting optimal parameters with random forest for quantum key distribution," *Quantum Information Processing*, 2023.

[100] A. Dhoha and et al., "Quantum cryptography on ibm qx," in *IEEE Conference Publication*, 2023.

[101] K. Yao, W. O. Krawec, and J. Zhu, "Quantum sampling for finite key rates in high dimensional quantum cryptography," *IEEE Transactions on Information Theory*, vol. 68, no. 5, pp. 3144–3163, 2022.

[102] L. Bacsardi, "On the way to quantum-based satellite communication," *IEEE Communications Magazine*, vol. 51, no. 8, pp. 50–55, 2013.

[103] W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned," *Nature*, vol. 299, no. 5886, pp. 802–803, 1982.

[104] C. H. Bennett and S. J. Wiesner, "Communication by epr devices," *Physical Review Letters*, vol. 69, no. 20, pp. 2881–2884, 1992.

[105] S. Goswami, S. Dhara, N. Sinclair, M. Mohageg, J. S. Sidhu, S. Mukhopadhyay, M. Krutzik, J. R. Lowell, D. K. L. Oi, M. Gundogan, Y.-C. Chen, H.-H. Jen, and C. Simon, "Satellites promise global-scale quantum networks," *arXiv preprint arXiv:2505.06693*, 2025.

[106] B. Author, "Quantum satellite communication," *arXiv preprint arXiv:2101.04427*, 2021.

[107] C. Author, "Terahertz quantum communication implementation," in *IEEE Conference*, 2019.

[108] C. H. Bennett and G. Brassard, "Quantum teleportation," *Phys. Rev. Lett.*, vol. 70, pp. 1895–1899, 1993.

[109] "6g communications overview." `https://example.com/6G-reference`, 2023.

[110] "Leo satellite quantum communication." `https://example.com/leo-satellites`, 2022.

[111] "Quantum algorithms for network optimization." `https://example.com/quantum-algorithms`, 2023.

[112] D. Author, "Quantum-aided multi-user transmission," in *IEEE Conference*, 2020.

[113] E. Author, "Quantum internet functionalities," *PubMed*, 2018.

[114] F. Author, "Underwater quantum communication," in *IEEE Conference*, 2016.

[115] G. Author, "Polarization fidelity in underwater qkd," *arXiv preprint arXiv:1207.6345*, 2012.

[116] Sitender, "A survey of machine learning and quantum computing for 5g/6g communications systems," *NeuroQuantology*, vol. 20, no. 22, pp. 1032–1043, 2022.

[117] H. Author, "Quantum computing and ai: Impacts & possibilities," *PDF*, 2022.

[118] A. Haldorai, S. Ramu, and D. Murugan, "Artificial intelligence and machine learning for future urban development," in *Computing and Communication Systems in Urban Development*, pp. 91–113, Cham: Springer, 2019.

[119] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, and J. R. McClean, "Power of data in quantum machine learning," *Nature Communications*, vol. 12, no. 1, pp. 1–9, 2021.

[120] S. R. Hasan, M. Z. Chowdhury, M. Sayem, and Y. M. Jang, "Quantum communication systems: Vision, protocols, applications, and challenges," *IEEE Access*, vol. 11, pp. 15855–15877, 2023.

[121] L.-M. Duan, "Quantum repeaters for long-distance communication," *arXiv preprint arXiv:2212.10820*, 2022.

[122] I. Author, "Improving entanglement distribution," in *IEEE Conference*, 2023.

[123] J. Author, "Coherence times in quantum memories," *Quantum Inf. Process.*, 2019.

[124] X. Wang, "Coexistence of discrete-variable qkd with wdm classical signals." `https://www.researchgate.net/publication/335932491_Coexistence_of_Discrete-Variable_QKD_with_WDM_Classical_Signals_in_the_C-Band_for_Fiber_Access_Environments`, 2019.

[125] K. Author, "Post-quantum cryptography: Mathematical foundations and future challenges." `https://www.researchgate.net/publication/393998738_Post-Quantum_Cryptography_Mathematical_Foundations_and_Future_Challenges`, 2023.

[126] L. Author, "Classical-quantum hybrid cryptography," *arXiv preprint arXiv:2411.01086*, 2024.

[127] M. Author, "Quantum error correction and noise management," *arXiv preprint arXiv:2210.00921*, 2022.

[128] V. Scarani, H. Bechmann-Pasquinucci, N. Cerf, M. Dusek, N. Lütkenhaus, and M. Peev, "The security of practical quantum key distribution," *Rev. Mod. Phys.*, vol. 81, pp. 1301–1350, 2009.

[129] H.-K. Lo, M. Curty, and K. Tamaki, "Secure quantum key distribution," *Nat. Photon.*, vol. 8, pp. 595–604, 2014.

[130] E. Diamanti, H.-K. Lo, B. Qi, and Z. Yuan, "Practical challenges in quantum key distribution," *npj Quantum Inf.*, vol. 2, p. 16025, 2016.

[131] M. Sasaki, M. Fujiwara, H. Ishizuka, and et al., "Field test of quantum key distribution in the tokyo qkd network," *Opt. Express*, vol. 19, pp. 10387–10409, 2011.

[132] R. Alléaume, C. Branciard, J. Bouda, and et al., "Using quantum key distribution for securing communications in existing infrastructures," *Quantum Inf. Process.*, vol. 13, pp. 1059–1083, 2014.

[133] R. Van Meter and J. Touch, *Designing Quantum Networks*. Wiley, 2013.

[134] H. A. Al-Mohammed and E. Yaacoub, "On the use of quantum communications for securing iot devices in the 6g era," in *2021 IEEE International Conference on Communications Workshops (ICC Workshops)*, pp. 1–6, IEEE, 2021.